

Cover Letter

To: Editorial Office, *npj Quantum Information* (Nature Portfolio, Springer Nature) **Date:** 25 May 2026 **From:** Aleaddin Özer (Corresponding Author), Chief Information Officer, E2E Solutions info@e2esolutions.tech · ORCID 0000-0001-9389-5357

Dear Editors,

Please find attached our manuscript, *“Three Axes of Quantum Risk: A Unified Observability Model for PQC, QKD, and Crypto-Agility,”* which we are submitting as an Original Research Article to *npj Quantum Information*.

The post-quantum transition is now an operational problem rather than a research-only one. NIST has finalised FIPS 203, 204, and 205; the U.S. NSA’s CNSA 2.0 timetable mandates broad migration between 2030 and 2035; and a parallel hardware track — Quantum Key Distribution governed by ETSI GS QKD 014 — offers an orthogonal channel-level control on high-assurance links. Yet the tools operators use to inventory their cryptographic posture continue to answer a single question: *“is this asset PQ-ready?”* That framing collapses two assets with very different migration costs into one bucket and is silent on the channel through which their key material travels.

This manuscript treats quantum-risk posture as a three-axis problem — algorithmic resistance (A), channel protection (C), and migration agility (G) — and combines the three with the operator’s deadline horizon into a single deadline-adjusted score $q(\text{asset}, t)$. The contribution is in three layers:

1. A formal three-axis posture model with explicit scoring rubrics, a deadline-tension term that shrinks the agility weight as the migration deadline approaches, and a worked four-asset example.
2. An open observability platform (Sezar) implementing the model end to end — eBPF-based TLS observation, an ETSI GS QKD 014 collector with a reusable Key-Management-Entity emulator, a static crypto-agility scanner, and a unified posture rollup. All under MIT at <https://github.com/e2esolutions-tech/sezar>.
3. Three reproducible empirical studies establishing a first baseline on real corpora — a Tranco-top-1k TLS-handshake survey (43.8% hybrid PQ-KEM adoption on 317 of 724 responsive hosts), a controlled ETSI 014 emulator study

characterising 13 induced KME/link failure modes (13/13 classification correctness), and a crypto-agility pilot with 91% agreement against hand-graded ground truth on an eleven-project subset of a fifty-project corpus.

The work sits within *npj Quantum Information*'s scope on three counts. The QKD axis treats channel-level quantum-secure key delivery as a first-class observable alongside the algorithmic PQC track, rather than as a separate concern — directly engaging quantum-information science's deployment frontier. The reproducibility of all three studies on commodity Linux without QKD hardware addresses the journal's emphasis on open, verifiable research. And the cross-disciplinary framing — security engineering meeting quantum information — fits Nature Portfolio's interest in work that draws practitioners from adjacent fields into the quantum-information conversation.

Submission disclosures

- **Originality.** The work has not been published or submitted elsewhere.
- **Authorship.** All authors have read and approved this submission. CRediT contributions are stated in the manuscript.
- **Competing interests.** None declared. E2E Solutions develops the Sezar reference implementation released under the MIT License but derives no direct commercial revenue from its publication.
- **Funding.** Self-funded by E2E Solutions and Hacettepe University; no external grants supported this work.
- **Data and code availability.** A repository-wide release at <https://github.com/e2esolutions-tech/sezar> contains the schema, the implementation, the ETSI 014 emulator and replay corpus, the Semgrep agility ruleset, the hand-graded ground-truth corpus, and the Tranco snapshot identifier used for Study 1.
- **Suggested reviewers.** Five names are listed in the accompanying suggested-reviewers.md file, covering PQ-TLS deployment (Stebila, Wiggers, Durumeric), QKD network engineering (Pacher), and crypto-agility / IETF standards (Housley). Three of the five are first authors on cited references. None has a co-author, advisor-student, or recent-grant relationship with either submitting author.
- **Licence.** Subject to acceptance, we will publish under CC-BY 4.0.

We thank the editors and the assigned reviewers for their consideration.

Sincerely,

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